Econ 101A
Section 9: Midterm Review (Solutions)

Clotaire Boyer*

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This is a review, not new material. The goal is to see the whole unit as one connected picture and to close out a few problems we did not finish in section. Nothing here is meant to be hard. If you can follow the map below and redo the short problems, you are in good shape. Take a breath, you know more than you think.

1 The Map: One Consumer Problem, Many Faces

Everything this unit fits on one page.

1. Preferences. Rational means *complete* (you can compare any two bundles) and *transitive* (no cycles). A utility function is just a numbering of bundles that respects the ranking. Any strictly increasing transform of u (like e^u or $\ln u$) represents the same preferences and gives the same demands.

2. Utility maximization. Choose the best affordable bundle:

$$\max_{x,y} u(x,y) \quad \text{s.t.} \quad px + qy = M, \qquad \mathcal{L} = u + \lambda(M - px - qy).$$

Interior optimum: tangency $u_x/u_y = p/q$, plus the budget. Watch for the traps: perfect substitutes and other corners, and non-convex preferences where the tangency is a minimum (check a corner or the second-order condition).

3. Marshallian demand and indirect utility. Solving gives $x^*(p,q,M)$ and $v(p,q,M) = u(x^*,y^*)$. By the envelope theorem, $\partial v/\partial M = \lambda$ (marginal utility of income) and $\partial v/\partial p = -\lambda x^*$.

4. The dual: expenditure minimization. Same tangency, different constraint:

$$\min px + qy \quad \text{s.t.} \quad u(x,y) \geq \bar{u} \;\Rightarrow\; \text{Hicksian } h(p,q,\bar{u}), \quad e(p,q,\bar{u}) = ph_x + qh_y.$$

Shephard: $\partial e/\partial p = h_x$. Duality: $h_x(p,q,\bar{u}) = x^*(p,q,e)$.

5. Slutsky. The total price effect splits into a substitution part (always ≤ 0) and an income part:

$$\frac{\partial x^*}{\partial p} = \left.\frac{\partial h_x}{\partial p}\right|_{\bar{u}=v} - x^* \frac{\partial x^*}{\partial M}.$$

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Normal good: both terms negative, demand slopes down. Giffen: needs an inferior good with an income effect large enough to flip the total positive. Every Giffen good is inferior; not every inferior good is Giffen.

6. Applications. Same tools, relabeled. Labor supply: leisure priced at the wage w, full income $M + wT$, tangency $u_\ell/u_c = w$, reservation wage below which hours are zero. Intertemporal: future consumption priced at $1/(1+r)$, wealth $W = M_0 + M_1/(1+r)$, Euler $U'(c_0) = \frac{1+r}{1+\delta}U'(c_1)$, smoothing when $r = \delta$.

2 Warm-up Problems

Problem 1 (preferences and transforms)

A consumer's preferences are represented by $u(x, y)$. Does $\widetilde{u} = e^u$ represent the same preferences? Would maximizing $\widetilde{u}$ subject to the same budget change the optimal bundle?

Solution. Take any two bundles a and b. Because the exponential function is strictly increasing, applying it to both sides of an inequality preserves it:

$$u(a) \geq u(b) \iff e^{u(a)} \geq e^{u(b)}.$$

So $\widetilde{u} = e^u$ ranks every pair of bundles exactly as u does. Same ranking means same preferences. Since the consumer's choice depends only on how bundles are ranked, not on the actual utility numbers, the optimal bundle is unchanged. (Contrast a *decreasing* transform such as $-u$: it flips every inequality, reverses the ranking, and does not represent the same preferences.)

Problem 2 (perfect substitutes)

Let $u(x, y) = 2x + y$ with prices $p, q > 0$ and income M. Find the optimal bundle and say when it is not unique.

Solution. Marginal utilities: $u_x = 2$, $u_y = 1$, so the MRS magnitude is $u_x/u_y = 2$, a constant. Interpret it as willingness to pay: the consumer will always trade 1 unit of y for $\frac{1}{2}$ unit of x, whatever the bundle. Compare this constant to the market rate p/q.

Rewrite the comparison as "bang per buck." Spending a dollar on x buys $1/p$ units, worth $2/p$ in utility; a dollar on y buys $1/q$ units, worth $1/q$. So x is the better buy when $2/p > 1/q$, i.e. $2q > p$, i.e. $p/q < 2$.

Case $p/q < 2$: put all income on x: $x^* = M/p$, $y^* = 0$.
Case $p/q > 2$: put all income on y: $x^* = 0$, $y^* = M/q$.
Case $p/q = 2$: the two goods give exactly the same utility per dollar, the indifference lines are parallel to the budget line, and *any* bundle on the budget line is optimal. That is the non-unique case.

Problem 3 (Cobb–Douglas quick pass)

Let $u(x, y) = \sqrt{xy}$, prices p, q, income M.

(a) State the Marshallian demands.

(b) Is x normal? Is it a gross substitute or complement for y?

Solution. (a) Derive them so the steps are visible. Write $u = (xy)^{1/2}$. Marginal utilities (chain rule):

$$u_x = \tfrac{1}{2}(xy)^{-1/2} \cdot y = \frac{y}{2\sqrt{xy}}, \qquad u_y = \tfrac{1}{2}(xy)^{-1/2} \cdot x = \frac{x}{2\sqrt{xy}}.$$

Tangency $u_x/u_y = p/q$: the $\frac{1}{2}$ and $\sqrt{xy}$ cancel, leaving $\dfrac{y}{x} = \dfrac{p}{q}$, i.e. $qy = px$. So the two goods get *equal expenditure*. Put $qy = px$ into the budget $px + qy = M$: $px + px = M$, so $2px = M$ and

$$x^* = \frac{M}{2p}, \qquad y^* = \frac{M}{2q}.$$

(b) Normality: $\dfrac{\partial x^*}{\partial M} = \dfrac{1}{2p} > 0$, so x is normal. Cross-price: $x^* = M/(2p)$ contains no q, so $\dfrac{\partial x^*}{\partial q} = 0$. With demand independent of the other price, x is neither a gross substitute nor a gross complement for y. (This is special to Cobb–Douglas.)

Problem 4 (Giffen versus inferior)

Using the Slutsky equation, explain why a Giffen good must be inferior, and why an inferior good need not be Giffen.

Solution. Write Slutsky for the own price:

$$\frac{\partial x^*}{\partial p} = \underbrace{\left.\frac{\partial h_x}{\partial p}\right|_{\bar{u}=v}}_{\text{substitution}} - \underbrace{x^* \frac{\partial x^*}{\partial M}}_{\text{income}}.$$

Step 1: the substitution term is always ≤ 0 (compensated law of demand: hold utility fixed, a higher own price shifts you away from x).
Step 2: "Giffen" means the total is positive, $\partial x^*/\partial p > 0$. Since the substitution term cannot be positive, the only way to get a positive total is for the income part $-x^*\,\partial x^*/\partial M$ to be positive and big enough to overturn it.
Step 3: with $x^* > 0$, that term is positive only if $\partial x^*/\partial M < 0$, i.e. the good is inferior. So Giffen forces inferior.
Step 4: the converse fails. An inferior good has $\partial x^*/\partial M < 0$, but if the income effect is small it will not overturn the negative substitution term, so the total stays negative and demand slopes down normally. Inferiority is necessary for Giffen, not sufficient.

Problem 5 (numerical Slutsky)

Keep $u = \sqrt{xy}$ at $p = q = 1$, $M = 8$, where $x^* = M/2p$, $y^* = M/2q$, and the Hicksian is $h_x = \bar{u}\sqrt{q/p}$. Decompose $\partial x^*/\partial p$ into substitution and income parts.

Solution. Initial point. $x^* = \dfrac{M}{2p} = \dfrac{8}{2} = 4$, $y^* = \dfrac{M}{2q} = 4$, and utility $u_0 = \sqrt{x^* y^*} = \sqrt{16} = 4$.

Substitution term $\partial h_x/\partial p$ at fixed $\bar{u} = u_0$. Write $h_x = \bar{u}\, q^{1/2} p^{-1/2}$. Differentiate in p: $\dfrac{\partial h_x}{\partial p} = \bar{u}\, q^{1/2} \cdot \left(-\tfrac{1}{2}\right) p^{-3/2} = -\dfrac{\bar{u}\sqrt{q}}{2p^{3/2}}$. Plug $\bar{u} = 4$, $p = q = 1$:

$$\frac{\partial h_x}{\partial p} = -\frac{4 \cdot 1}{2 \cdot 1} = -2.$$

Income term $-x^* \, \partial x^*/\partial M$. From $x^* = M/(2p)$, $\dfrac{\partial x^*}{\partial M} = \dfrac{1}{2p} = \dfrac{1}{2}$. So $-x^* \dfrac{\partial x^*}{\partial M} = -4 \cdot \dfrac{1}{2} = -2$.

Add them. $\dfrac{\partial x^*}{\partial p} = -2 + (-2) = -4$.

Check directly. From $x^* = M/(2p)$, $\dfrac{\partial x^*}{\partial p} = -\dfrac{M}{2p^2} = -\dfrac{8}{2} = -4$. It matches. Both pieces are negative and x is normal, so x is an ordinary (downward-sloping) good, not Giffen.

Problem 6 (applications one-liners)

(a) In the labor model with non-labor income M and endowment T, what is the reservation wage, and what happens below it?

(b) In the two-period model with $r = \delta$, what is the relation between c_0 and c_1, and how do you know whether the consumer saves?

Solution. (a) The reservation wage is the wage at which desired hours first turn positive. Below it, the interior formula would ask for more leisure than exists, so the consumer sits at the corner $\ell = T$, $h = 0$, $c = M$. (With $u = \ln c + \ln \ell$ we found it equals $w = M/T$.)
(b) With $r = \delta$ the Euler equation gives $U'(c_0) = U'(c_1)$; since U' is strictly decreasing, equal marginal utilities force $c_0 = c_1$ (perfect smoothing). She saves when income is front-loaded, i.e. $M_0 > c_0$, moving resources into the leaner period. Saving is $s = M_0 - c_0$.

One last thing. If a problem looks unfamiliar, do not panic. Write the budget, write the Lagrangian, take first-order conditions, and read off what the question asks. The steps are always the same. Good luck, you have got this.